HW 4 — Jackson Eq. 7.90

black = write it down · red = the answer · teal = my note, do not copy · grey box = optional

Problem — General solution of the 1D wave equation

> Derive **Jackson Eq. 7.90**, the general solution of the one-dimensional wave equation.

What to write: Steps 0 → 3, ending at Eq. 7.90. That is the whole answer. Everything after the dashed line on page 4 is background, not required.

ANSWER

$$u(x,t) = \frac{1}{2}\,\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dk\; A(k)\, e^{i[kx - \omega(k)t]} + \text{c.c.}$$

*c.c. = complex conjugate of the term before it. A(k) is a **free function** — that is what makes the solution general.*

Where each step comes from in the question:

"wave equation"	Step 0
"one-dimensional" → $\nabla^2 \to \partial^2/\partial x^2$	Step 1
"**general**" → all k, not one wave	Step 2
u is physical → real → the "+ c.c."	Step 3

Step 0 — Starting point

Maxwell in vacuum gives the wave equation:

$$\nabla^2 \vec{E} - \mu_0\varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} = 0, \qquad \mu_0\varepsilon_0 = \frac{1}{c^2}$$ Lec 1, slide 13

Each component obeys it separately. Take E_x, expand in a Fourier integral:

$$E_x(\vec{x},t) = \frac{1}{(2\pi)^{3/2}} \int d^3k\; e^{i\vec{k}\cdot\vec{x}}\; \varepsilon_x(\vec{k},t)$$ Lec 1, slide 16

*3D → 1D: $d^3k \to dk$, $(2\pi)^{3/2} \to \sqrt{2\pi}$. ε_x still depends on t — it is a(k,t) of Step 1, **not** Jackson's A(k). Result also stated in* Lec 8, slide 9

Step 1 — One dimension; time dependence of each k

Write $u = E_x$. Jackson uses the name u because every component obeys the same scalar equation — the derivation does not care which field it is.

In 1D, $\nabla^2 = \partial^2/\partial x^2$:

$$\frac{\partial^2 u}{\partial x^2} - \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = 0$$

Write u as a **Fourier integral**:

$$u(x,t) = \frac{1}{\sqrt{2\pi}} \int dk \; e^{ikx}\, a(k,t)$$

Here **a(k,t)** = how much of the wave e^{ikx} is present in u at time t.

$\partial^2/\partial x^2$ gives $(ik)^2 = -k^2$. The integrand must vanish for each k:

$$\frac{\partial^2 a}{\partial t^2} + k^2 c^2 a = 0 \quad \textit{(Helmholtz)}$$ Lec 1, slide 18

Its two solutions are $e^{-i\omega t}$ and $e^{+i\omega t}$:

$$a(k,t) = C(k)\, e^{-i\omega t} + D(k)\, e^{+i\omega t}, \qquad \omega = c|k|$$ Lec 1, slide 18

C and D are the two constants of integration, fixed in Step 3.

> *Slide 18 writes $\omega = kc$, where $k = |\vec{k}|$ is a 3D magnitude, never negative. In 1D k is signed, so $\omega = c|k|$ — even in k, as needed next.*

$\omega = c|k|$ is the **vacuum** answer. In glass or water the speed depends on frequency, so the curve is different. Jackson therefore writes $\omega = \omega(k)$, meaning "whatever function of k it happens to be", so that 7.90 holds in any medium:

$$\omega = \omega(k), \quad \omega(-k) = \omega(k) \qquad \textit{Jackson 7.79}$$

The second condition says a wave going left has the same frequency as the same wave going right. It is used in Step 3, where k is flipped to −k.

Step 2 — Superposition over all k

Put a(k,t) of Step 1 into the Fourier integral of Step 1:

$$u(x,t) = \frac{1}{\sqrt{2\pi}} \int dk \; e^{ikx} \left[C(k)\, e^{-i\omega t} + D(k)\, e^{+i\omega t} \right]$$

Keep the $e^{-i\omega t}$ term only and rename its coefficient $C(k) \to A(k)$:

$$u(x,t) = \frac{1}{\sqrt{2\pi}} \int dk \; e^{ikx} A(k)\, e^{-i\omega t}$$

The two exponentials multiply: $e^{ikx} e^{-i\omega t} = e^{i(kx - \omega t)}$, so

$$u(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk \; A(k)\, e^{i[kx - \omega(k)t]}$$ Jackson 7.80

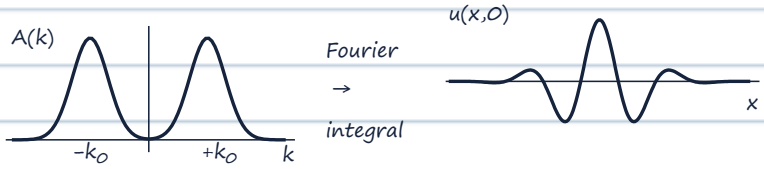
The D-term is not lost — it returns on its own in Step 3 as the "+ c.c."

It is **general** because:

① the equation is **linear** → any sum of solutions is a solution;

② A(k) is free for every k.

One k = an endless sine wave. A band of k added together = a pulse:



Narrow band in k → long pulse in x; wide band → short pulse.

*A(k) is a new symbol, **not** C(k) — their relation comes in Step 3. The $1/\sqrt{2\pi}$ is just the Fourier convention.*

Step 3 — Make u real → Eq. 7.90

u is physical, so it must be real. For any complex z:

$$\mathrm{Re}\, z = \frac{1}{2} \left(z + z^* \right)$$

Apply that to 7.80:

FINAL ANSWER — Jackson 7.90

$$u(x,t) = \frac{1}{2}\,\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dk\; A(k)\, e^{i[kx - \omega(k)t]} + \text{c.c.}$$

*A(k) stays as a free function — that is what makes the solution **general**. Fixing A(k) would pick out one particular wave.*

■ THIS IS THE ANSWER — STOP COPYING HERE ■

EXTRA — not part of the answer

1. Where the D-branch went. Conjugate the integral, then put $k \to -k$:

$$\text{c.c.} = \frac{1}{2}\,\frac{1}{\sqrt{2\pi}}\int dk\; A^*(-k)\, e^{i[kx + \omega(k)t]} \quad \textit{(uses } \omega(-k) = \omega(k)\textit{)}$$

That is the $e^{+i\omega t}$ term of Step 1. Comparing with a(k,t):

$$C(k) = \frac{1}{2}A(k), \qquad D(k) = \frac{1}{2}A^*(-k)$$

So C and D are not independent — 7.90 carries only one unknown function.

EXTRA — not part of the answer

2. What A(k) is, if you are given a starting wave (Jackson 7.91).

2nd order in t needs two conditions: u(x,0) and ∂u/∂t(x,0). Put t = 0 in 7.90, then differentiate 7.90 and put t = 0:

$$u(x,0) = \frac{1}{2}\,\frac{1}{\sqrt{2\pi}}\int dk\ [\,A(k) + A^*(-k)\,]\,e^{ikx}$$

$$\frac{\partial u}{\partial t}(x,0) = \frac{1}{2}\,\frac{1}{\sqrt{2\pi}}\int dk\ (-i\omega)[\,A(k) - A^*(-k)\,]\,e^{ikx}$$

Invert both, then add them — $A^*(-k)$ cancels:

$$A(k) = \frac{1}{\sqrt{2\pi}}\int dx\ e^{-ikx}\left[\,u(x,0) + \frac{i}{\omega(k)}\,\frac{\partial u}{\partial t}(x,0)\,\right]$$

Jackson 7.91

3. The lecture's own notation. No ½ in front — it sits inside A:

$$u(x,t) = \frac{1}{\sqrt{2\pi}}\int dk\ A(k)\,e^{i[kx - \omega t]} + \text{c.c.}, \quad A_{lec} = \frac{1}{2}A_{Jackson}$$ Lec 8, slide 9

OPTIONAL **Check.** With $\omega = c|k|$, split at k = 0: k > 0 gives ik(x−ct), k < 0 gives ik(x+ct), so u = f(x−ct) + g(x+ct) — d'Alembert. ✓ Also checked numerically: u came out real and both initial conditions were reproduced to 11 decimals.